SECTION-I

Single Correct Answer Type **13 Q. [3 M (–1)]**

1. One requires 11eV of energy to dissociate a carbon monoxide molecule into carbon and oxygen atoms. The minimum frequency of the appropriate electromagnetic radiation to achieve the dissociation lies in
(A) visible region (B) infrared region (C) ultraviolet region (D) microwave region

2. A linearly polarized electromagnetic wave given as $\vec{E} = E_0\hat{i}\cos(kz-\omega t)$ is incident normally on a perfectly reflecting infinite wall at z = 0. Assuming that the material of the wall is optically inactive, the reflected wave will be given as

(A) $\vec{E}_r = -E_0\hat{i}\cos(kz-\omega t)$ (B) $\vec{E}_r = E_0\hat{i}\cos(kz+\omega t)$

(C) $\vec{E}_r = -E_0\hat{i}\cos(kz+\omega t)$ (D) $\vec{E}_r = E_0\hat{i}\sin(kz-\omega t)$

3. The electric field intensity produced by the radiations coming from 100 W bulb at a 3m distance is E. The electric field intensity produced by the radiations coming from 50 W bulb at the same distance is

(A) $\frac{E}{2}$ (B) 2E. (C) $\frac{E}{\sqrt{2}}$ (D) $\sqrt{2}E$

4. Electromagnetic wave consists of periodically oscillating electric and corresponding magnetic vectors .....
(A) in randomly oriented planes but vibrating in phase.
(B) in mutually perpendicular planes but vibrating in phase.
(C) in mutually perpendicular planes but vibrating with a phase difference of π.
(D) in mutually perpendicular planes but vibrating with a phase difference of π/2.

5. A flood light is covered with a filter that transmits red light. The electric field of the emerging beam is represented by a sinusoidal plane wave $E_x = 36 \sin(1.20\times10^7 z - 3.6\times10^{15} t)$ V/m
The average intensity of the beam will be-
(A) 0.86 W/m^2 (B) 1.72 W/m^2 (C) 3.44 W/m^2 (D) 6.88 W/m^2

6. An electromagnetic wave of frequency 1 x 10^{14} hertz is propagating along z-axis. The amplitude of electric field is 4 V/m. If $\varepsilon_0 = 8.8\times10^{-12}\frac{C^2}{N-m^2}$, then average energy density of electric field will be:

(A) 35.2×10^{-10} J/m^3 (B) 35.2×10^{-11} J/m^3

(C) 35.2×10^{-12} J/m^3 (D) 35.2×10^{-13} J/m^3

7. A plane electromagnetic wave of angular frequency ω propagates in a poorly conducting medium of conductivity σ and relative permittivity ε. Find the ratio of conduction current density and displacement current density in the medium.

(A) $\frac{\varepsilon\varepsilon_0\omega}{\sigma}$ (B) $\frac{\sigma}{\varepsilon\varepsilon_0\omega}$ (C) $\frac{\omega}{\sigma\varepsilon_0\varepsilon}$ (D) $\frac{\omega\sigma}{\varepsilon_0\varepsilon}$

8. The electric field of a plane polarized electromagnetic wave in free space at time t = 0 is given by an expression $\vec{E}(x,y) = 10\hat{j}\cos[(6x+8z)]$ The magnetic field $\vec{B}$ (x, z, t) is given by : (c is the velocity of light)

(A) $\frac{1}{c}(6\hat{k}+8\hat{i})\cos[(6x-8z+10ct)]$ (B) $\frac{1}{c}(6\hat{k}-8\hat{i})\cos[(6x+8z-10ct)]$

(C) $\frac{1}{c}(6\hat{k}+8\hat{i})\cos[(6x+8z-10ct)]$ (D) $\frac{1}{c}(6\hat{k}-8\hat{i})\cos[(6x+8z+10ct)]$

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9. An observer is standing 90 cm away from a point source of light of power 250W. RMS value of electric field due to the source at the position of the observer is :-
(A) 48V/m (B) 96V/m (C) 40V/m (D)80V/m

10. The Sun is directly overhead and you are facing toward the north. Light coming to your eyes from the sky just above the horizon is :-
(A) Partially polarized north-south (B) Partially polarized east,west
(C) Partially polarized up-down (D) Randomly polarized

11. The angle of incidence of light is equal to Brewster's angle, then
(a) reflected ray is perpendicular to refracted ray
(b) refracted ray is parallel to reflected ray
(c) reflected light is polarized having its magnetic field vector in the plane of incidence
(d) refracted light is polarized
(A) (a) and (d) are true (B) (a) and (b) are true
(C) (a) and (c) are true (D) (b) and (c) are true

12. The resolving power of a lens can be improved by increasing :
(A) Diameter of the lens (B) The object distance
(C) The wavelength of light (D) The brightness of light

13. In single slit fraunhoffer diffraction pattern, resultant ampitude at the centre of central maxima is A_0 and it is A at an angluar position of 30º. Then $\frac{A}{A_0}$ will be - [Slit width a = λ]

(A) $\frac{\pi}{2}$ (B) $\frac{2}{\pi}$ (C) $\frac{\pi}{4}$ (D) $\frac{4}{\pi}$

Multiple Correct Answer Type 3 Q. [4 M (–1)]

14. An electromagnetic wave travelling along z-axis is given as: $\vec{E} = E_0 \cos(kz - \omega t)$. Choose the correct options from the following;

(A) The associated magnetic field is given as $\vec{B} = \frac{1}{c}\vec{k} \times \vec{E} = \frac{1}{\omega}(\vec{k} \times \vec{E})$

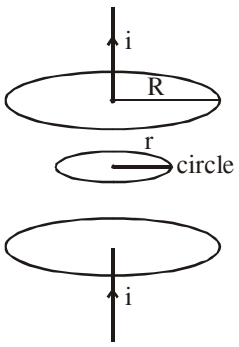
(B) The electromagnetic field can be written in terms of the associated magnetic field as $\vec{E} = c(\vec{B} \times \hat{k})$

(C) $\hat{k}.\vec{E} = 0, \hat{k}.\vec{B} = 0$

(D) $\hat{k} \times \vec{E} = 0, \hat{k} \times \vec{B} = 0$

15. A plane electromagnetic wave propagating along x direction can have the following pairs of **E** and **B**
(A) E_x, B_y. (B) E_y, Bz. (C) B_x, E_y. (d) E_z, B_y.

16. A capacitor with circular plates of radius R is charged with a constant current I as shown. Which of the following are correct?



(A) Poynting vector points radially inward everywhere on the circle.
(B) Magnetic field is clockwise as seen from above.
(C) Magnetic energy densisty is increasing with time.
(D) Poynting vector flux is equal to the rate of change of stored energy.

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Linked Comprehension Type (1 Para × 2Q.) [3 M (-1)]
(Single Correct Answer Type)

Paragraph for Question No. 17 and 18

It is known that accelerating charges radiate electromagnetic waves. This is the method of power being radiated by a cellphone. A cellphone has an antenna in which we can assume that charges undergo oscillation with an amplitude x0 and angular frequency ω. This average power can be assumed to depend on charge Q, x_0, ω, t_0 and velocity of light c.

17. The formula for power P is (a is dimensionless constant) :-

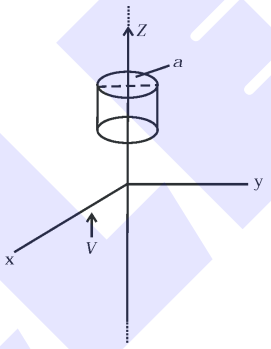
(A) $\frac{aq^2\omega^4x_0^2}{\epsilon_0 c^2}$ (B) $\frac{aq^2\omega^4x_0^2}{\epsilon_0 c^3}$ (C) $\frac{aq^4\omega^2x_0}{\epsilon_0^2 c^6}$ (D) $\frac{aq^4\omega^4x_0^2}{\epsilon_0^2 c^4}$

18. A cell phone radiates 3W power at 300 MHz. If a = $\frac{1}{3}$, find the number of photons being emitted per unit time ?

(A) 1.5×10^{25} /sec (B) 8×10^{23}/sec (C) 4×10^{19} /sec (D) Information insufficient

Subjective Type 11 Q. [4 M (0)]

19. An infinitely long thin wire carrying a uniform linear static charge density λ is placed along the z-axis figure. The wire is set into motion along its length with a uniform velocity $\vec{v} = v\hat{k}_z$. Calculate the pointing vector at point (a, 0, 0)



20. A long straight cable of length l is placed symmetrically along z-axis and has radius a(<<l). The cable consists of a thin wire and a co-axial conducting tube. An alternating current $I(t) = I_o \sin(2\pi vt)$ flows down the central thin wire and returns along the co-axial conducting tube. The induced electric field at a distance s from the wire inside the cable is $\vec{E}(s,t) = \mu_0 I_0 v \cos(2\pi vt) \ln\left(\frac{s}{a}\right)\hat{k}$

(i) Calculate the displacement current density inside the cable.
(ii) Integrate the displacement current density across the cross-section of the cable to find the total displacement current I_d.
(iii) Compare the conduction current I_0 with the displacement current I_d.

21. A plane electromagnetic wave propagates in a homogeneous and isotropic medium with ε = 3.00 and μ = 1.00. The amplitude of the wave's electric field strength is E_m = 10.0 V/m. Find:
(a) the amplitude of the wave's magnetic field strength H_m.
(b) the phase speed v of the wave.

22. A plane electromagnetic wave with ω of the order of 10^{10} s^{-1} propagates in a vacuum. The amplitude of the wave's electric vector is E_m = 0.775 V/m. A wave-absorbing surface having the shape of hemisphere of radius r = 0.632 m is placed in the wave's path with its vertex facing the direction of propagation of the wave. What energy W will this surface absorb during the time t = 1.00 sec.

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23. A parallel-palte capacitor with roudn plates is charged with steady current during the time τ to a voltage of U. The separation distance is d. Imagining that there is a cylindrical surface between the plates coaxial to them whose radius r is much smaller than the radius of the plates, determine:
(a) the magnitude and direction of the Poynting vector at points of the surface;
(b) the amount of energy W flowing through the surface during the time t. Compare W with the energy of the electric field contained in the volume V confined by the surface after completion of the charging process.

24. The current in a very long solenoid increases uniformly form zero to I during the time t. The number of solenoid turns per unit length is n. Imagining inside the solenoid in its middle part a closed surface coaxial to it of length ℓ and radius r, determine:
(a) the magnitude and direction of the Poynting vector at points of the surface;
(b) the amount of energy W flowing through the surface during the time t. Compare W with the energy of the magnetic field contained in the volume V confined by the surface after the current I has set in.

25. A plane electromagnetic wave propagate along the x-axis in vacuum. The maplitude of the wave's magnetic field strength is $H_m = 0.0500$ A/m. Determine:
(a) the amplitude of the wave's electric field strength E_m;
(b) the time-averaged density <w> of the wave's energy;

26. Magnitude of average poynting vector at the surface of sun is $n\times10^7 \frac{W}{m^2}$. Find n. (Given that power radiated by sun is 3.8×10^{26} W and radius of sun is 7×10^8 m)

27. Earth recieves $2\frac{cal}{min\ cm^2}$ solar energy. What is the amplitude of electric field at Earth's surface?

28. An electromagnetic wave with a frequency of v = 3.0 MHz passes from vacuum to a non-magnetic medium with a dielectric constant $\varepsilon_r = 4.0$. Find the change in its wavelength.

29. A leaky capacitor with circular plates of radius R = 12 cm has a capacitance of 5 μF and an effective resistance of $4 \times 10^5\ \Omega$. At t = 0, the potential difference across the plates is zero but is increasing at the constant rate of 2000 V/s. (a) Find the displacement current I_D. (b) At what time is I_D equal to the conduction current?

GR # DIFFRACTION, POLARIZATION AND EM WAVES — ANSWER KEY

SECTION-I

Single Correct Answer Type — **13 Q. [3 M (–1)]**

1. Ans. (C)	**2. Ans. (B)**	**3. Ans. (C)**	**4. Ans. (B)**	**5. Ans. (B)**
6. Ans. (C)	**7. Ans. (B)**	**8. Ans. (B)**	**9. Ans. (B)**	**10. Ans. (B)**
11. Ans. (C)	**12. Ans. (A)**	**13. Ans. (B)**		

Multiple Correct Answer Type — **3 Q. [4 M (–1)]**

14. Ans. (A,B,C)	**15. Ans. (B,D)**	**16. Ans. (A,D)**

Linked Comprehension Type (1 Para × 2Q.) — **[3 M (-1)]**

(Single Correct Answer Type)

17. Ans. (B)	**18. Ans. (A)**

Subjective Type — **11 Q. [4 M (0)]**

19. Ans. $\frac{\lambda^2 v}{4\pi^2\varepsilon_0 a^2}\hat{k}$ **20. Ans. (i)** $\frac{2\pi}{\lambda^2} I_0 \ln\left(\frac{a}{s}\right)\sin(2\pi vt)\hat{k}$, **(ii)** $+\left(\frac{a}{2}\right)^2\left(\frac{2\pi}{\lambda}\right)^2 I_0 \sin 2\pi vt$, **(iii)** $\frac{I_0^d}{I_0} = \left(\frac{a\pi}{\lambda}\right)^2$

21. Ans. (a) 46 mA/m, (b) $\sqrt{3} \times 10^8$ **m/s** **22. Ans. 1 mJ**

23. Ans. (a) $\frac{r\varepsilon_0 u^2}{2\tau^2 d^2}t$, **(b)** $\frac{\varepsilon_0 u^2 \pi r^2}{2d}$ **24. Ans. (a)** $\frac{\mu_0 n^2 I^2 r}{2\tau^2}t$; **(b)** $\frac{\mu_0 n^2 I^2 \pi r^2 \ell}{2}$

25. Ans. (a) 18.84 V/m; (b) 1.57×10^{-9} **26. Ans. 6.17** **27. Ans. 1026.94**

28. Ans. 50 **29. Ans. (a) 0.01 A; (b) 2 s**

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